

# Depth over Fidelity

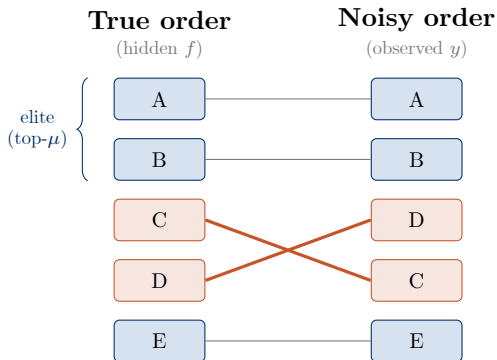
## in Fixed-Budget Noisy Evolution Strategies

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# The Problem: Misranking



- CMA-ES uses only the **ranking** of candidates.
- Noise flips ranks — a **misranking**.
- The wrong candidate enters the elite set, **corrupting the update**.

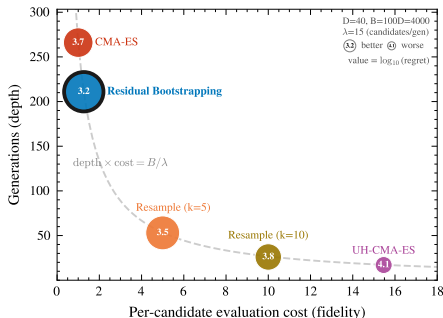
# The Trade-off: Fidelity Costs Depth

- Each generation samples  $\lambda$  candidates; resampling each  $k$  times denoises the ranking — higher **fidelity**.
- But under a fixed budget, depth shrinks:

$$T \approx \frac{B}{k\lambda}.$$

( $\lambda$  = population size)

- So fidelity is bought with **depth** — the number of update steps.

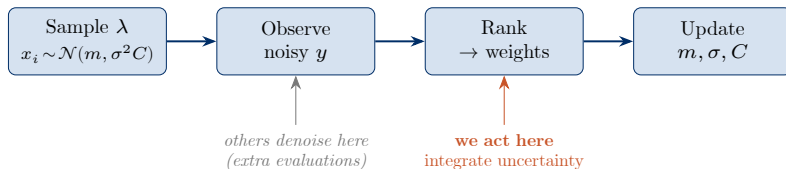


# Depth over Fidelity

When the ranking is noisy,  
don't pay to clean it up.

**Absorb the uncertainty into the update  
— and keep iterating.**

# Where We Intervene

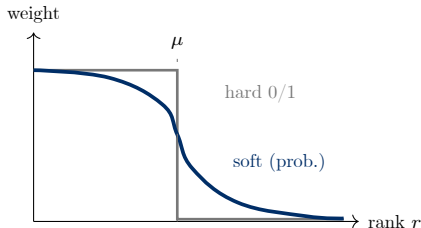


$$m^+ = m + \eta \sum_i w(r_i) (x_i - m)$$

CMA-ES mean update — PEM will replace the weights  $w(r_i)$

Keep a **one-evaluation baseline** — with only small capped overhead, we change how ranking becomes weights  $w(r_i)$ .

# PEM: Probabilistic Elite Membership



- Replace the hard elite label by a **probability**:

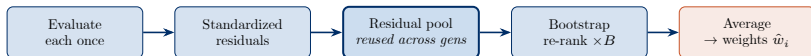
$$w_i^* = \Pr(r_i \leq \mu \mid x_{1:\lambda}).$$

- A borderline candidate gets a *fractional vote*.
- A **Rao–Blackwellization** of the noisy step:

$$\Delta m_{\text{PEM}} = \mathbb{E}[\Delta m^{\text{noisy}} \mid x_{1:\lambda}]$$

same expected step, **less variance**.

# RB-PEM: Residual Bootstrapping

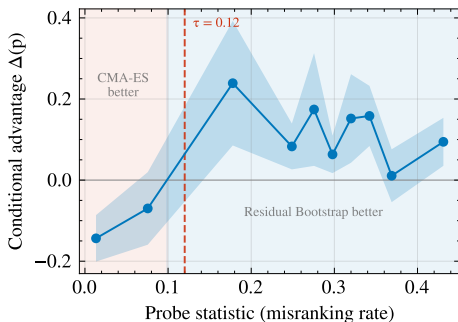


$$\hat{w}_i = \frac{1}{B} \sum_{b=1}^B w(\text{rank}_i(\tilde{y}^{(b)})), \quad \tilde{y}^{(b)} = \hat{f}(x) + \hat{s}(x) z^{(b)}, \quad z^{(b)} \sim \widehat{D}$$

synthetic values from the fitted noise model  $\widehat{D}$  — re-ranked, then averaged

Bootstrap rankings are **software-only**; targeted re-evaluation is capped at  $K_{\max} \leq 1$ , so depth stays near  $B/\lambda$ .

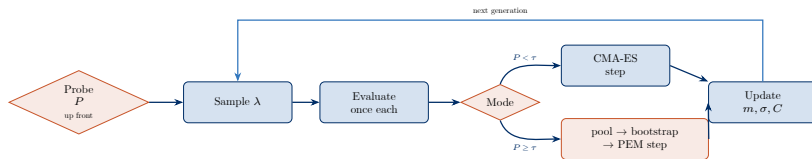
# Probe-and-Switch



- A cheap **probe**  $P$  measures rank instability.
- $P \geq \tau \Rightarrow$  run **RB-PEM**.
- $P < \tau \Rightarrow$  fall back to **CMA-ES**.
- **Robust**: gains where they exist; reverts when ranks are stable.



# The Full Algorithm



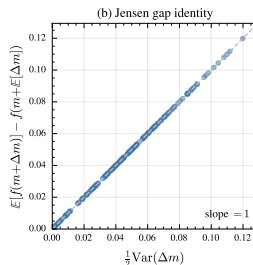
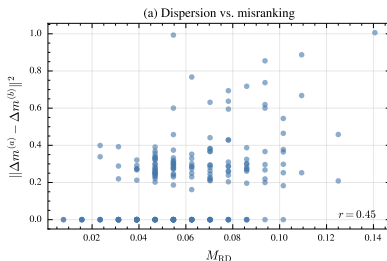
Probe once, then use a one-evaluation baseline plus capped overhead when active — **depth is near-maximal**.

# Why It Works: The Curvature Tax

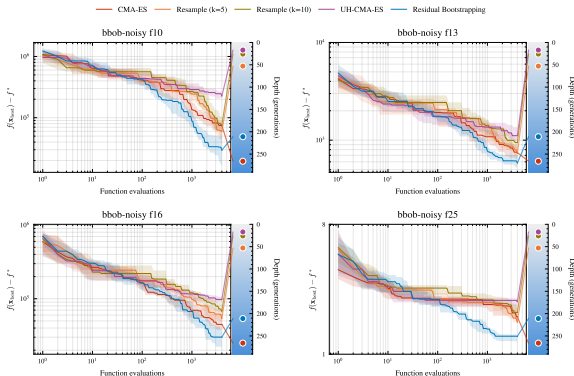
$$\mathbb{E}[f(X) \mid x_{1:\lambda}] \geq f(\bar{X}) + \frac{\alpha}{2} \mathbb{E}[\|X - \bar{X}\|^2 \mid x_{1:\lambda}]$$

Theorem 1 (proof in paper)    $\alpha$ : local curvature    $\|X - \bar{X}\|^2$ : update dispersion

Update **dispersion**  $\times$  curvature = an unavoidable **per-generation tax**. PEM minimizes the variance, so it minimizes the tax.



# Evidence



Probe-and-Switch win rate (COCO + 6 external tasks):

| vs Res.(10)  | vs Res.(5)   | vs UH-CMA-ES | vs CMA-ES |
|--------------|--------------|--------------|-----------|
| <b>85.6%</b> | <b>79.3%</b> | <b>77.4%</b> | 54.3%     |

Ranks strongest on high-misranking functions, ahead of LRA-CMA-ES; the CMA-ES edge grows with dimension.

# Takeaway

## Depth over Fidelity

- Under noisy rankings + tight budgets, **integrate** uncertainty into selection rather than spending evaluations to remove it.
- Largest gains where misranking is high; *robust* elsewhere.
- Portable to other population-based optimizers.

Thank you!

[github.com/sichen-wang/Depth-over-Fidelity\\_ICML2026](https://github.com/sichen-wang/Depth-over-Fidelity_ICML2026)



Code